PRODUCTION AND OPERATIONS MANAGEMENT

PRODUCTION AND OPERATIONS MANAGEMENT

POMS

Vol. 27, No. 3, March 2018, pp. 531–552
ISSN 1059-1478 | EISSN 1937-5956 | 18 | 2703 | 0531
DOI 10.1111/poms.12813

The Influence of Intellectual Property Rights on Poaching in Manufacturing Outsourcing

Keith Skowronski*
Darla Moore School of Business, University of South Carolina, 1014 Greene Street, Columbia, South Carolina 29208, USA, keith.skowronski@moore.sc.edu

W.C. Benton Jr.
Fisher College of Business, The Ohio State University, 2100 Neil Ave, Columbus, Ohio 43210, USA, benton.1@osu.edu

Manufacturers are often placing their intellectual property (IP) at risk by outsourcing to suppliers in countries with weak IP rights. Thus, understanding how to safeguard their IP from poaching, which is a supplier's unauthorized use of a buyer's proprietary information, is of critical practical importance for manufacturers that outsource to suppliers in countries with weak IP rights. To investigate this phenomenon, we use dyadic data from globally dispersed manufacturer–supplier relationships to examine how the IP rights of a supplier's location affects poaching and how IP rights influence the effectiveness of two transactional characteristics—supplier idiosyncratic investments and media-rich communication—on poaching. We examine these two transaction characteristics because they are representative of two forms of safeguards which create self-enforcing agreements that do not require legal enforcement—economic and relational. We find that the strength of IP rights not only has a direct effect on poaching, but it also influences the relationships for both transaction characteristics. Intriguingly, when IP rights are weak, we find that not only are supplier idiosyncratic investments less effective in reducing poaching, but increases in these investments are associated with an increase in poaching. This finding illustrates that the strength of IP rights is a determinant of the degree to which supplier idiosyncratic investments are transaction specific. For suppliers in countries with weak IP rights, media-rich communication is found to be more effective in reducing poaching. This finding illustrates the importance of manufacturers' boundary spanners in building relationships with suppliers in countries with weak IP rights.

Key words: opportunism; intellectual property rights; transaction specific assets; boundary spanning; offshore outsourcing
History: Received: December 2015; Accepted: October 2017 by Edward Anderson, after 3 revisions.

1. Introduction

A firm's ability to innovate is a key driver to creating a competitive advantage (Porter 1985), but a firm needs more than the ability to innovate. To profit from innovation, firms must have safeguards in place that protect their intellectual property (IP) (Teece 1986). With the rise of globalization, firms are often sourcing from suppliers in countries with weak IP rights (Commission on the Theft of American Intellectual Property 2013). Legal mechanisms that can protect IP in countries with strong IP rights, such as patents and contracts, can be ineffective at protecting a firm's IP in countries with weak IP rights (Chesbrough et al. 2006, Gassmann et al. 2012). Thus, when outsourcing to suppliers in countries with weak IP rights, a US manufacturer must utilize alternative methods to protect their IP, or a manufacturer could fall victim to IP theft by an offshore supplier.

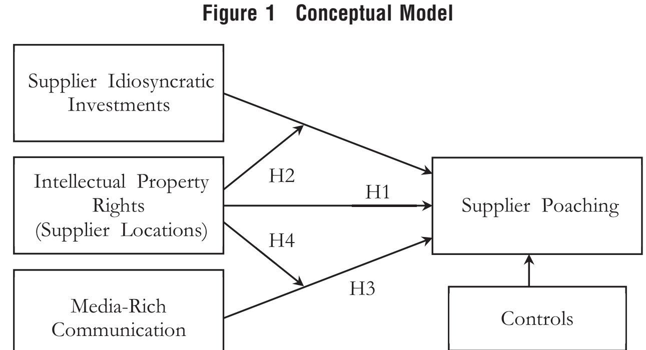
For instance, consider the following example, observed in practice, of a US-based discrete-products manufacturer that outsourced to an offshore supplier. The manufacturer, who had previously only outsourced to domestic suppliers, decided to outsource to offshore suppliers (i.e., suppliers in emerging markets) to lower production costs. The manufacturer managed the outsourcing arrangement similar to those with domestic suppliers. At the onset of these outsourcing engagements, the manufacturer highlighted the complexities of the processes required to produce the components being outsourced because the manufacturer had found that suppliers that were willing to invest time and resources in learning how to manufacture these products were committed to maintaining their relationship with the manufacturer. The manufacturer provided product specifications, such as drawings and computer-generated models, and a physical sample of the finished components. Additionally, to assist the suppliers in ramping up production, the manufacturer provided their standard operating procedures for the in-house operations. Once the supplier's production was up and running, the manufacturer's purchasing personnel managed the ongoing relationship. Domestic and offshore suppliers were managed similarly (e.g., utilized the same contracts, required similar investments into the relationships by domestic or offshore suppliers, etc.). The main difference between the relationship management practices utilized between domestic and offshore suppliers was that face-to-face and telephone communication were less frequent for offshore suppliers due to the cost of overseas travel and differences in working hours across time zones. Initially, the manufacturer was relatively pleased with the offshore supplier's performance. But the manufacturer's view of the offshore supplier drastically changed about one year into the relationship when one of the managers in the manufacturing firm discovered the manufacturer's proprietary designs in the supplier's catalog, listed with a part number and a price. At that point, it was apparent that the investments the supplier had made in learning how to manufacture these components led to the manufacturer's IP, specifically its designs and processes, being used to supply other customers. In this particular case, the manufacturer fell victim to supplier poaching, and the manufacturer determined they had little recourse due to the weak IP rights where the supplier was located.

Intellectual property theft by a supply chain partner is a form of opportunism called poaching (Aron et al. 2005). While the previous example may seem extreme, similar IP protection concerns have manifested themselves in buyer–supplier relationships, as manufacturers have been victim to poaching by suppliers in emerging markets. For example, Volkswagen (VW) found that one of its Chinese suppliers was supplying components produced using VW-owned tooling directly to VW's competitor SAIC Chery (Shengjun and Fernandez 2007). Similarly, Gray et al. (2017) interviewed a US manufacturer that outsourced to an offshore supplier and fell victim to supplier poaching. The US manufacturer discovered this at a trade show when replicas of their proprietary designs were displayed in a competitor's booth, and tooling marks on those products made it apparent that the manufacturer's offshore supplier produced them. Concerns of poaching in manufacturing supply chains have become serious enough that the United States International Trade Commission (USITC) warns manufacturers about suppliers in emerging markets running "midnight shifts" to create counterfeit products off of manufacturers' designs and tooling (USITC 2010). Thus, understanding how manufacturers can protect their IP when utilizing suppliers in weak IP rights locations is of critical practical importance.

When firms outsource to suppliers in emerging economies, they forgo two of the main protections for IP: vertical integration (Williamson 1991) and legal safeguards (Teece 1986). For this reason, opportunism is one of the main concerns when outsourcing (Holcomb and Hitt 2007, McIvor 2009), and IP theft, or poaching, is one of the main "hidden costs" that a firm can encounter when outsourcing manufacturing (Burton 2013). We examine this phenomenon in the context of manufacturing outsourcing by testing a model (see Figure 1) that examines two research questions: (1) How do IP rights affect supplier poaching? (2) How do IP rights affect the relationships between two transaction characteristics—supplier idiosyncratic investments and media-rich communication—and supplier poaching?

We examine these two transaction characteristics because they are representative of the two main forms of safeguards, economic and relational, which create self-enforcing agreements that do not require legal enforcement (Dyer and Singh 1998, Jap and Anderson 2003). Thus, these safeguards may protect manufacturers' IP in the absence of legal protections; a situation encountered when sourcing from suppliers in countries with weak IP rights. Idiosyncratic investments are investments in assets that are specialized to produce a specific good or service and have a lower value when not producing that good or service. Relationships where both parties have made investments in specialized assets (i.e., bilateral idiosyncratic investments) have been considered safeguarded from opportunism (Dyer and Singh 1998, Jap and Anderson 2003, Wathne and Heide 2000). However, others have argued that these same investments by a supplier can lead to increases in a specific form of opportunism, poaching, when IP rights are weak (Aron et al. 2005, Teece 1986). We empirically evaluate this argument to see if a supplier's investments into specialized assets, which have previously been considered a form of commitment to a relationship, have the opposite effect and lead to an *increase* in poaching for suppliers in countries with weak IP rights. We examine media-rich communication because, in the absence of strong legal institutions, firms often rely on relational governance (Liu et al. 2009, Luo 2007). Relational governance refers to the building of close

Figure 1 Conceptual Model



relationships between the buyer and supplier, which requires close, personal forms of communication (i.e., media-rich communication) that can be difficult to maintain for manufacturers with suppliers located offshore. We empirically evaluate these relationships utilizing a unique dyadic dataset of globally dispersed manufacturing outsourcing relationships that we combine with secondary data of the location characteristics for these dyadic relationships.

2. Theoretical Background

2.1. Opportunism in Buyer–Supplier Relationships

Opportunism is defined as "self-interest seeking with guile" (Williamson 1985, p. 47), where guile is defined as "lying, stealing, cheating, and calculated efforts to mislead, distort, disguise, obfuscate, or otherwise confuse" (Williamson 1985, p. 47). Opportunism can be classified into three distinct categories: forced renegotiation, shirking, and poaching (Aron et al. 2005). Forced renegotiation is the classic hold-up problem, or trying to extract *ex-post* higher rents (Klein et al. 1978). Shirking is the intentional underperformance of agreed upon duties (Alchian and Demsetz 1972). Poaching is the unauthorized use of the other party's IP for revenue streams outside of the current relationship (Aron et al. 2005). Of the previous studies examining the antecedents to opportunism in exchange relationships, only Handley and Benton (2012) and Handley and Angst (2015) conceptualized opportunism to include poaching. Of those two studies, only Handley and Benton (2012) examined the drivers of poaching separately from other forms of opportunism, as Handley and Angst (2015) conceptualized opportunism as a second-order construct containing poaching and shirking. In this study, we also examine poaching in isolation from other forms of opportunism because governance mechanisms that can be used to control shirking and forced renegotiation may not be effective in controlling poaching (Aron et al. 2005). This study differs substantially from that of Handley and Benton (2012) in two ways. First, this study examines IP rights, which is a key consideration in the security of IP (Williamson 1991). Second, we examine poaching in the context of manufacturing outsourcing. We chose this context, as opposed to the business process outsourcing context examined previously in Handley and Benton (2012), for several reasons: (1) manufacturers often outsource to suppliers in countries with weak IP rights (Rossetti and Choi 2005), (2) suppliers in manufacturing supply chains often have expertise similar to their manufacturing customers (i.e., the manufacturers), enabling the suppliers to become competitors of the manufacturers (Aron et al. 2005), and (3) when manufacturers outsource manufacturing activities, they provide suppliers with proprietary information that makes it easier for the supplier to imitate the technology (Teece 1986).

Supplier poaching can include actions such as counterfeiting and piracy, which involve the unauthorized reproduction of another's physical or digital IP, respectively (USITC 2011). This study also differs substantially from work in those areas. Studies on counterfeiting and piracy have examined why consumers choose counterfeits (e.g., Gao et al. 2017, Qian and Xie 2014), the effects of pirated products on firm revenue or brand reputation (e.g., Givon et al. 1995, Qian 2014), reactive strategies once counterfeits have appeared in the market (e.g., Anand and Galetovic 2004, Cho et al. 2015, Yang et al. 2004), and factors that inform the supplier selection process (e.g., Massimino et al. 2017). This study focuses on proactive supplier relationship management strategies that firms can implement to protect IP, which places this study closer to the literature examining how managers of foreign firms located in weak IP rights countries (e.g., the manager of a plant in China of a US-owned firm) prevent IP leakage that is reviewed in section 2.3. To examine how IP rights can affect poaching in manufacturing outsourcing, we draw from multiple literature streams, which are outlined in the following sections.

2.2. Transaction Cost Economics

Transaction cost economics (TCE) posits that the degree to which parties to a recurring transaction may be susceptible to opportunism is related to the degree to which each party has incurred investments in assets that are specialized to produce the product or service being exchanged (Williamson 1979). Since these assets are specialized, there are limited opportunities for use outside of the product or service for which they were designed to produce; thus, the return on these investments is highest when producing that specialized good or service. Additionally, since there can be a limited number of exchange partners for the specialized product or service, these assets can become specific to that transaction and, under those circumstances, are referred to as transaction specific assets. When a firm in an exchange relationship invests in highly transaction specific assets, it can create a lock-in situation, which arises because the firm could lose the value of their investments in these productive assets if that relationship were terminated. This situation opens the investing firm up to opportunistic acts by the other party in the relationship because the investing firm does not want to lose the value of their investment.

Transaction cost economics (TCE) refers to recurring transactions between autonomous entities that exhibit considerably more cooperation and coordination than found in a market transaction as bilateral governance (Williamson 1979, 1985) or hybrid modes of governance (Williamson 1991, 1996). To illustrate this type of arrangement, Williamson (1979) uses the example of a manufacturer purchasing a specialized component, which is similar to the manufacturing outsourcing context examined in this research. In discussing this example, Williamson (1979, p. 242) states:

> "Thus assume (a) that special-purpose equipment is needed to produce the component in question (which is to say that the value of the equipment in its next-best alternative use is much lower), (b) that scale economies require that a significant, discrete investment be made, and (c) that alternative buyers for such components are few (possibly because of the organization of the industry, possibly because of special-design features). The interests of buyer and seller in a continuing exchange relation are plainly strong under these circumstances."

Thus, recurring transactions to which both parties have made idiosyncratic investments create a lock-in situation for both parties. Williamson (1985) describes these offsetting idiosyncratic investments as safeguards, while other researchers have supported the notion that relationships are safeguarded from opportunism when both parties have made idiosyncratic investments (e.g., Dyer and Singh 1998, Jap and Anderson 2003, Wathne and Heide 2000). The current study fits nicely into this literature by examining if there are situations where a supplier's idiosyncratic investments do not safeguard a buyer from supplier opportunism. To examine this, we draw from the extant literature on the institutional environment and specifically focus on how IP rights affect the relationship between supplier idiosyncratic investments and opportunism.

2.3. Intellectual Property Rights

In his seminal work on profiting from innovation (PFI), Teece (1986) argues that protecting an innovation from imitation is contingent on the appropriability regime, which is the ease of imitation from both a legal and technological standpoint. Weak appropriability regimes are those where technology is more difficult to protect because of both low legal protections and technology that is easy to replicate. Thus, in the manufacturing outsourcing context where the supplier is already manufacturing the buyer's components, the appropriability regime is in essence solely determined by the strength of IP rights because the supplier already possesses the knowledge required to replicate the product. Specifically, Teece argues that firms are susceptible to imitation (i.e., poaching) by suppliers in weak IP rights locations when specialized assets (i.e., idiosyncratic investments) are required for the supplier to manufacture the product. Similarly, Aron et al. (2005) discuss how suppliers in locations with weak IP rights may invest heavily into specialized assets to learn as much as possible from the buyer and then redeploy those specialized assets through poaching. Thus, IP rights can affect the relationship between supplier idiosyncratic investments and a specific form of opportunism, poaching, and modify it from that of a safeguarding relationship when IP rights are strong to a relationship where supplier idiosyncratic investments exacerbate supplier opportunism when IP rights are weak.

The PFI framework also posits that firms can rely on contractual arrangements when technology is relatively easy to protect (i.e., a tight appropriability regime); however, in weak appropriability regimes, protecting an innovation requires combining technology with other assets that are difficult to replicate, referred to as complementary or co-specialized assets (Teece 1986). Similarly, Williamson (1991) argues that the strength of governance mechanisms is contingent on the institutional environment, specifically the IP rights where the operation will be located. Thus, how a firm protects its proprietary information depends on the strength of IP rights, and an emerging stream of research has examined the actions that managers at multinational enterprises (MNEs) in locations with weak IP rights take to protect their proprietary information. Consistent with our manufacturing outsourcing context, we review studies that can inform how firms safeguard operations that are being performed at suppliers in locations with weak IP rights, as opposed to studies that have examined characteristics of tasks performed in weak IP rights locations (e.g., Quan and Chesbrough 2010, Zhao 2006), other location-related characteristics that inform the supplier selection process (e.g., Massimino et al. 2017), or strategies that require the process to be performed in-house, such as secrecy (e.g., Lawson et al. 2012).

Recent studies have found that managers at MNEs operating in China can protect IP through relationship-building efforts. For example, Keupp et al. (2009, 2010) found that building relationships with both employees and external bodies (i.e., internal and external guanxi) can safeguard IP for foreign firms with operations in China. Similarly, Schotter and Teagarden (2014) posit that building relationships in China create social capital that can assist in IP protection. More broadly, Holmes et al. (2016) argue that embeddedness and building relationships with other firms become critical to safeguarding information in any weak IP rights location. These arguments are consistent with the PFI framework as relationships with suppliers are considered a complementary resource (Helfat and Lieberman 2002). Thus, developing relationships with suppliers can be a method to protect IP. Relatedly, the supply chain literature has shown that relational governance can guard against opportunism in locations with weak legal protections, which we discuss in the next section.

2.4. Relational Governance

Buyer–supplier relationships are repeated exchanges that are embedded in social interactions (Granovetter 1985). These repeated social interactions lead to shared norms and values that govern the relationship, which is broadly referred to as relational governance (Dyer and Singh 1998, Uzzi 1997). Exchanges utilizing relational governance differ from market transactions by exhibiting cooperation, fine-grained information transfer, and joint-problem solving (Dwyer et al. 1987, Macneil 1978). Thus, relational governance requires communication modes that facilitate social interactions and iterative problem-solving processes, which are communication modes high in media richness (Dyer and Singh 1998). Daft and Lengel (1986) categorize communication modes on a continuum of media richness, which refers to the amount of information transmitted and the ability to handle ambiguity. Communication modes high in media richness (i.e., media-rich communication), such as face-to-face and telephone communication, transmit more fine-grained information than those with less media richness, such as written communication, and communication modes high in media richness are better for situations that contain ambiguity, such as joint-problem solving.

While media-rich communication is also required to create human asset specificity (Dyer 1996), for firms utilizing relational governance, media-rich communication encompasses more than just specialized assets. For example, Uzzi (1997, p. 46) found that this type of communication "is also more than a matter of asset-specific know-how or reducing information asymmetry between parties, because the social relationship imbues information with veracity and meaning beyond its face value." Researchers in other contexts have also highlighted the importance of media-rich communication in building relationships. Orlikowski (2002) found that face-to-face communication was required to build and maintain social networks in geographically distributed product development. Similarly, He and Nickerson (2006, p. 47) observed that in the trucking industry "carriers develop friendships with shippers and frequently call them on the telephone to maintain their relationships." Levina and Vaast (2005) observed that face-to-face communication was critical in building the social capital required to be an effective inter-organizational boundary spanner. Furthermore, Levina and Vaast (2008) illustrated that these activities become even more important when firms are geographically dispersed, as firms encounter country-level and time-zone differences. Thus, media-rich communication is instrumental in building inter-organizational relationships.

Engaging in media-rich communication becomes more challenging with offshore suppliers (Levina and Vaast 2008, Parker and Anderson 2002). As geographic distance increases, so do the costs associated with face-to-face communication. Additionally, operating in different time zones can create difficulties in communicating by telephone. These complications can inhibit relationship-building efforts with offshore suppliers. For example, Levina and Vaast (2008) found that travel budgets constrained managers in distributed development projects, making it difficult to form relationships with their offshore developers. Additionally, examples from Gray et al. (2017) illustrate these complications in manufacturing outsourcing. They interviewed managers at manufacturers that rarely visited their offshore suppliers after outsourcing, treating these exchanges closer to arm's length transactions when compared to the manufacturers' typical exchanges with domestic suppliers. They found that offshore relationships that exhibited less media-rich communication resulted in reduced control over the exchange, which resulted in unanticipated problems in these relationships, such as opportunism by the offshore suppliers. In one instance, the manufacturer even fell victim to IP theft by the offshore supplier. Overcoming these complications with offshore suppliers is critical as relational governance can be a safeguard from opportunism, especially in the absence of legal protections (Liu et al. 2009, Wang et al. 2013, Zhou and Xu 2012). Thus, utilizing media-rich communication should safeguard outsourcing relationships from poaching, and that safeguarding relationship should be stronger for suppliers in locations with weak IP rights.

2.5 Summary

There are several unexplored relationships in the literature that we examine in this study. First, TCE posits that a supplier's investments in specialized assets should make the supplier less likely to act opportunistically (i.e., poach), but the relationships underlying TCE assume strong IP rights. Conversely, literature on outsourcing and IP rights (e.g., Aron et al. 2005, Teece 1986) argues that, under the condition of weak IP rights, a supplier that invests in specialized assets may be more likely to poach (i.e., act opportunistically). In this study, we contribute to both literature streams by examining whether the effect of a supplier's idiosyncratic investments on poaching is contingent on the strength of IP rights. We also contribute to the literature on relational governance by examining the effects of media-rich communication. Although TCE argues that engaging in media-rich communication creates human asset specificity, literature on relational governance (e.g., Dyer and Singh 1998, Uzzi 1997) argues that exchanges utilizing higher levels of media-rich communication (i.e., above that required for specialized assets) exhibit relational governance. Thus, the relationships formed through media-rich communication should act as a safeguard from opportunism, which becomes increasingly more important in the absence of legal safeguards. We examine these arguments by testing whether media-rich communication serves as a safeguard for poaching while controlling for supplier idiosyncratic investments and whether the strength of that relationship is contingent on the strength of IP rights.

3. Hypotheses

3.1. Intellectual Property Rights

Strong IP rights are a mechanism to deter supplier poaching. Deterrence theory posits that increases in the severity of the punishment associated with an action reduce the occurrence of that action (Paternoster 2010). IP rights determine the severity of the legal sanctions associated with unauthorized use of proprietary information. Suppliers in countries with strong IP rights have legal disincentives (i.e., punishments) for poaching that are not present in countries with weak IP rights. Thus, under weak IP rights, a supplier can appropriate a buying firm's knowledge with diminished fear of legal sanctions (Teece 1986, Williamson 1991). For that reason, suppliers in countries with weak IP rights are more likely to use proprietary information gained from the buyer for revenues outside of that relationship. We formally hypothesize:

Hypothesis 1. *The strength of the intellectual property rights in the supplier's location is negatively associated with supplier poaching.*

3.2. Supplier Idiosyncratic Investments and Intellectual Property Rights

One of the transaction characteristics that affects opportunism in buyer–supplier relationships is the degree to which either party has made investments into specialized human assets (i.e., those attained from learning by doing) and physical assets (Williamson 1996). Since these assets are specialized, they are limited in their use outside of the product or service for which they were designed to produce. Thus, the return on these investments is highest when producing that particular specialized good or service. The specialized nature of these investments creates a small numbers bargaining situation (i.e., few possible exchange partners) and creates an *ex-post* dependency in the relationship. Larger idiosyncratic investments can increase the dependency on that relationship because the firm will forgo the value of the idiosyncratic investment if that relationship is terminated. This makes these investments a method for buying firms to reduce supplier opportunism (i.e., a safeguard), since a supplier that has invested heavily in a relationship will not want to act in a way that could jeopardize the relationship (Jap and Anderson 2003, Wathne and Heide 2000). For the supplier, these investments create a "lock-in" situation with the buyer because the supplier has limited options outside of the current relationship. For that reason, idiosyncratic investments have been considered a credible commitment between exchange partners (Anderson and Weitz 1989, Williamson 1983). Higher levels of supplier idiosyncratic investments have also been shown to reduce supplier shirking and forced renegotiation (Achrol and Gundlach 1999, Gundlach et al. 1995, Katsikeas et al. 2009, Liu et al. 2009). A similar relationship should exist between supplier idiosyncratic investments and poaching for suppliers located in countries with strong IP rights. When IP rights are strong, a supplier cannot use proprietary knowledge gained through that exchange relationship outside of that relationship. Thus, in these countries, if a supplier poaches and the buyer terminates the relationship, the supplier will incur the loss of those investments. The supplier investment loss should serve as a disincentive for poaching, with higher investments being stronger disincentives.

Transaction cost economics posits that increases in supplier idiosyncratic investments will lead to a reduction in supplier opportunism because these investments can create an *ex-post* small numbers bargaining situation for the supplier. In these situations, the buyer may be the only exchange partner with whom the supplier can maximize their return on the idiosyncratic investment; therefore, the assets are considered transaction specific. As Williamson (1996) points out, the relationships described in TCE assume the presence of a western-style institutional environment, such as strong IP rights, and these relationships will be affected by changes in the institutional environment. Under strong IP rights, a supplier's idiosyncratic investments become specific to the transaction because the buyer is the only exchange partner that can purchase the good or service from the supplier. For example, if a supplier is manufacturing a product that the buyer designed (i.e., the buyer's IP), the supplier can only manufacture that product for the buyer because the supplier could incur legal sanctions if it attempted to recreate and sell that product to other customers. The condition of weak property rights can alter this *ex-post* small numbers bargaining situation. Under the condition of weak IP rights, the buyer is often not the only potential customer for that good or service, as the supplier can offer that product or service to the market due to the reduced fear of legal sanctions. As Aron et al. (2005, p. 50) discuss:

> "A [supplier] may, over time, become a competitor of its [customer]. This is more frequently observed in manufacturing outsourcing, where a [supplier] reverse-engineers designs and begins to act as a direct competitor. Numerous examples are well known, from stereo production to chip manufacturing."

Similarly, Teece (1986) argues that, under the condition of weak IP rights, specialized assets enable a supplier to become a competitor of the buyer. Thus, a supplier attempting to maximize the return on its investments in specialized assets might engage in opportunism (i.e., poaching) to maximize that return, and larger idiosyncratic investments create a larger incentive to engage in poaching. In essence, weak IP rights introduce additional avenues for a supplier to gain returns on idiosyncratic investments, such as becoming the buyer's competitor (Aron et al. 2005, Rossetti and Choi 2005, Teece 1986) or even possibly counterfeiting (Midler 2009, USITC 2010). A supplier in a country with weak IP rights may be motivated to use their idiosyncratic investments outside of the relationship because the supplier can gain additional revenue streams due to the reduced fear of legal sanctions. Put simply, increases in supplier idiosyncratic investments by suppliers in weak IP rights countries may lead to increases in supplier poaching.

To summarize, when IP rights are strong, a supplier's idiosyncratic investments should have a safeguarding effect on supplier poaching, similar to the relationships observed between these investments and other forms of opportunism. On the other hand, when IP rights are weak, specialized assets may lead to increases in poaching. Thus, the relationship between supplier idiosyncratic investments and supplier poaching is contingent on the level of IP rights. Formally, we hypothesize:

HYPOTHESIS 2. *Intellectual property rights negatively moderate the relationship between supplier idiosyncratic investments and supplier poaching.*

3.3. Media-Rich Communication and Intellectual Property Rights

Idiosyncratic investments are economic incentives that safeguard a party from opportunism, but safeguards can also be relational in nature (Dyer and Singh 1998). Specifically, building a close relationship with an exchange partner, or relational governance, is another method that can be utilized to safeguard transactions (Dwyer et al. 1987, Macneil 1985). Relational governance refers to repeated exchanges that are embedded in social interactions, which lead to shared norms and values that govern the relationship (Granovetter 1985). Thus, engaging in relational governance requires repeated personal interactions between parties in an exchange. To build a relationship, parties must engage in communication modes that foster personal contact, which are those modes high in media richness. Research across multiple contexts has shown that the building and maintaining of relationships requires boundary spanners to engage in communication modes high in media richness, such as face-to-face (Levina and Vaast 2005, Orlikowski 2002) and telephone communication (He and Nickerson 2006). Relational governance is also characterized by increased cooperation and joint-problem solving (Dwyer et al. 1987, Macneil 1978), which are fostered by media rich modes of communication because they facilitate rapid feedback and more easily address ambiguity (Dyer and Singh 1998). Thus, media-rich communication is required to engage in relational governance. Relational governance has been shown to lead to positive outcomes in exchange relationships in studies with samples from mature economies (e.g., Brown et al. 2000, Poppo and Zenger 2002, Prahinski and Benton 2004) and emerging economies (e.g., Liu et al. 2009, Zhou and Xu 2012). Since media-rich communication has been shown to foster better relationships between firms, we expect that outsourcing relationships managed by boundary spanners that engage in media rich modes of communication should see similar benefits, regardless of the strength of IP rights. Specifically, relational safeguards have been found to be negatively associated with opportunism in both mature markets (e.g., Handley and Angst 2015, Tangpong et al. 2010) and emerging markets (e.g., Liu et al. 2009, Luo 2007). Thus, we expect a similar safeguarding relationship between media-rich communication and a specific form of opportunism, poaching. Formally, we hypothesize:

HYPOTHESIS 3. *Higher levels of media-rich communication are associated with a decreased risk of supplier poaching.*

While we expect that higher levels of media-rich communication should lead to a reduction in poaching independent of the location of the supplier, this safeguarding relationship should be stronger for manufacturers with suppliers in countries with weak

IP rights. When a manufacturer outsources to suppliers in countries with strong IP rights, there are multiple ways to safeguard IP, such as patents, contracts, or economic safeguards, such as bilateral idiosyncratic investments, that are not effective when outsourcing to suppliers in weak IP rights locations. By definition, IP protection mechanisms such as patents are ineffective in countries with weak IP rights. Furthermore, "contracts are seldom sufficient in developing contexts" (Chesbrough et al. 2006, p. 57), and, as posited in Hypothesis 2, supplier idiosyncratic investments could lead to an increase in poaching. Thus, when engaging with suppliers in weak IP rights locations, relational governance is one of the few governance mechanisms available to a manufacturer. For that reason, actions that contribute to building a relationship, such as engaging in media rich modes of communication, should be of increased importance when dealing with suppliers in these locations.

Research across multiple contexts supports the increased importance of relational governance, and media-rich communication specifically, in countries with weak IP rights. For example, Levina and Vaast (2008) illustrated that engaging in media-rich communication by boundary spanners becomes more important when building relationships with offshore suppliers. Relatedly, Luo (2007) found that close boundary spanner ties lead to a reduction in opportunism in international joint ventures. Zhou and Xu (2012) found that relational governance curbed local supplier opportunism for multinational enterprises operating in China. Similarly, Keupp et al. (2009, 2010) found that building relationships with both employees and external bodies can safeguard IP for foreign firms operating in China. Finally, others have posited that embeddedness and social relationships can protect IP in any weak property rights locations (Holmes et al. 2016, Schotter and Teagarden 2014). Thus, boundary spanners engaging in media-rich communication should be even more important in safeguarding a manufacturer from poaching when managing suppliers in countries with weak IP rights. Formally, we hypothesize:

HYPOTHESIS 4. *The negative relationship between media-rich communication and supplier poaching is stronger when IP rights are weak.*

4. Research Methods

4.1. Data Collection and Sample Characteristics

The unit of analysis for this study is a manufacturer (buyer)–supplier dyad. To obtain dyadic responses, we contacted manufacturers and asked participants to respond about a specific supplier to whom the manufacturer had outsourced the manufacturing of products. The respondents answered questions regarding the overall relationship between the two firms, including listing all of the buyer plant locations that received products from the supplier. In the buyer survey, we requested the name and e-mail address for the respondent's key contact at the supplier, which was subsequently used to survey the supplier. The supplier respondents also answered questions about the total relationship between the two firms, including listing all of the supplier plant locations that manufactured products for the buyer.

4.1.1. Pilot Study. The survey instrument was developed using an iterative process in the second and third quarters of 2014. The scales were obtained through an extensive literature review. Industry experts then reviewed the survey instrument, and their feedback was used to refine several items. A pilot study was then conducted in the fourth quarter of 2014 to refine the survey administration procedures (Forza 2002). The member list of a supply chain continuous improvement organization that consisted of supply chain practitioners was used to contact potential respondents for the pilot study. Initial and follow-up contacts were sent through e-mail, and participants were offered an executive summary of the findings as an incentive for participation. A total of 27 responses were received for a 5.8% response rate, and all respondents provided supplier contact information. Suppliers were then contacted using a similar procedure. Eight responses were received for a 28% supplier response rate. After the pilot study, the data collection procedure was modified following Dillman et al. (2009) to include a different method of contact, specifically follow-up by telephone, to increase the response rate.

4.1.2. Data Collection. The list of potential buyer respondents for the large-scale data collection was obtained from a business-to-business contact information provider. Since we were interested in manufacturing outsourcing relationships, we limited the population to manufacturers in discrete product industries, specifically focusing on firms in NAICS industries 332 through 339. We included manufacturers with 2013 revenues exceeding \$10 million USD because poaching is a problem faced by both small-to-medium sized enterprises (SMEs) and large firms (USITC 2010). Potential respondents were identified as supply chain professionals (i.e., manager level or higher) because they should have detailed knowledge of the total relationship between the two firms. We initially contacted each potential respondent with an e-mail explaining the purpose of the study and benefits of participation. The first follow-up contact was made by telephone, and two subsequent follow-ups were sent through e-mail. The researchers were unable to identify some of the target respondents online or by telephone and concluded that these potential respondents had been inappropriately identified for participation in the survey. In total, 888 target respondents were assumed not to be qualified to be a participant due to reasons including change of jobs, retirement, or just no sufficient involvement in manufacturing outsourcing. Of the remaining 3132 target respondents, 256 respondents returned a questionnaire for an effective response rate of 8.2%. Subsequently, 164 of those that responded provided supplier contact information. Using the same administration procedure as we did for the buyer survey, we contacted the 164 suppliers and received 101 responses for a 62% response rate. Non-response bias was examined using archival analysis (Rogelberg and Stanton 2007). No significant differences were found between respondents and non-respondents on t-tests examining the differences in revenue ($p > 0.20$) and employees ($p > 0.20$). These 101 responses were combined with the eight responses from the pilot study. Five observations were dropped due to missing data for single-item control variables, making the sample size for the analysis 104 matched dyads. The sample characteristics are shown in Table 1.

Table 1 Sample Descriptive Statistics

Buyer respondent title		Buyer firm size (rev.)		Supplier respondent title	
Sr. VP	2%	\$10–\$24.9 million	34%	CXO	12%
VP	5%	\$25–\$49.9 million	15%	Sr. VP	6%
Sr. Director	5%	\$50–\$99.9 million	17%	VP	11%
Director	13%	\$100–\$249.9 million	13%	Sr. Director	2%
Sr. Manager	18%	\$250–\$499.9 million	5%	Director	10%
Manager	49%	\$500–\$999.9 million	5%	Sr. Manager	18%
Other	8%	\$1–\$4.9 billion	8%	Manager	22%
		> \$5 billion	3%	Other	20%

Buyer manufacturing industry	
Fabricated Metal Product	4%
Machinery	36%
Computer and Electronic Product	13%
Electrical Equipment, Appliance, and Component	7%
Transportation Equipment	22%
Furniture and Related Product	6%
Medical Equipment and Miscellaneous	13%

4.2. Measures

All scales for the survey instrument were adapted from previous studies whenever possible, with specific changes made for the manufacturing context. Since research has found that exchange partners differ in the way they view aspects of the relationships (Roh et al. 2013), we asked the best-informed party for constructs where one party may have limited access to information. For example, since the supplier should be better informed on the level of the supplier's idiosyncratic investments than the buyer, that construct was measured on the supplier survey. When both parties have similar access to information (e.g., media-rich communication frequency), we asked the buyer to reduce concerns of common method bias due to the independent and dependent variables coming from the same source (Podsakoff et al. 2003). The survey questions used for the multi-item constructs in the main analysis can be found in Appendix A.

The dependent variable, *poaching*, is taken from the supplier survey and measures the extent that the supplier might engage in using potentially proprietary information from the buyer for gains outside of that relationship using a multi-item construct from Handley and Benton (2012). *Supplier idiosyncratic investments* are a measure of supplier investments in specialized physical and human assets, measured on the supplier survey with a multi-item scale adapted from Anderson and Weitz (1992). *Media-rich communication* measures the frequency of communication modes high in media richness, with the scales drawn from communication modes in Carlson and George (2004), and was measured on the buyer survey.

The strength of IP rights was operationalized using a country-level secondary measure of *IP rights* from the International Property Rights Index (IPRI 2014). The IPRI has been published annually and is comprised of three separate indexes: (1) the legal and political environment, (2) physical property rights, and (3) IP rights. We use the *IP rights* measure in the main analysis but examine the IPRI measure as a robustness check. The respondents completing the supplier survey were asked to list all of the supplier plant locations where products for the buyer are manufactured. The level of *IP rights* for the supplier was then calculated by averaging the *IP rights* scores for all of the supplier's plant locations, giving a supplier *IP rights* score for the total relationship. The IPRI overall composite measure and all three of its components are measured on a 0–10 scale, with 10 being the strongest property rights. For 2014, values for the *IP rights* measure ranged from 2.6 to 8.6. For our sample, the *IP rights* scores range from 5.4 to 8.4. The average *IP rights* value for our sample is 7.69 with a standard deviation of 1.15. Thirty percent of our sample contains suppliers with plant locations in weak IP rights countries as categorized by Zhao (2006).

4.2.1. Covariates. Several control variables that could influence both the dependent and independent variables were also included in this study. Two covariates were included to control for legal mechanisms to limit poaching—*patent* and *detailed contracts*. A binary variable from the buyer survey, *patent*, was coded "1" if the buyer had any patents associated with the outsourced products. The measure of *detailed contracts* was captured on the buyer survey using a multi-item measure from Zhou and Xu (2012).

Five transaction characteristics that could affect the relationships between *supplier idiosyncratic investments*, *media-rich communication*, and *poaching* were also included: *average distance*, *sole source*, *outsourcing degree*, *relationship length*, and *electronic information sharing*. *Average distance* captures the average distance between plants in the buyer and supplier networks in kilometers, calculated using the Haversine formula, and was included because distance can affect both communication and monitoring. *Sole source* was measured on the buyer survey and coded "1" if the supplier is the sole source, "2" for dual sourcing, and "3" for more than two suppliers for these products. *Sole source* was included because idiosyncratic investments have been shown to increase as the number of suppliers decreases (Aral et al. 2017). *Outsourcing degree* measures the amount of manufacturing the buyer performs internally. It was measured on the buyer instrument and coded "1" if the buyer has multiple stages of production performed in-house, "2" if only assembly is performed in-house, and "3" if no manufacturing is done internally. *Outsourcing degree* was included because as the buyer performs less manufacturing internally, supplier idiosyncratic investments and communication should increase. *Relationship length*, which was measured in years and taken from the buyer instrument, was included to account for the differences between newly formed and long-term relationships. Additionally, *electronic information sharing* was included to control for the frequency that the firms electronically share information because it could affect the frequency of media-rich communication, and it was measured using a multi-item construct from Zhou and Benton (2007) on the buyer survey instrument.

Two measures were included to control for potential future revenues that the supplier could obtain from the relationship (i.e., a supplier's economic dependence): *buyer size* and *supplier revenue*. *Buyer size* is a measure of buyer firm size that was obtained with the respondent information (1 = \$10–\$24.9 million; 2 = \$25–\$49.9; 3 = \$50–\$99.9; 4 = \$100–\$249.9; 5 = \$250–\$499.9; 6 = \$500–\$999.9; 7 = \$1–\$4.9 billion; 8 = ≥ \$5 billion). *Supplier revenue* is a measure of the revenue that the supplier generates from the buyer (1 = < \$1.0 million; 2 = \$1.0–\$24.9; 3 = \$25.0–\$49.9; 4 = \$50.0–\$99.9; 5 = \$100–\$499.9; 6 = ≥ \$500.0) and was measured on the supplier survey.

Three additional covariates were included to control for the opportunity to poach: *buyer idiosyncratic investments*, *product customization*, and *new products*. *Buyer idiosyncratic investments* are measured on the buyer survey with a multi-item construct from Anderson and Weitz (1992). While both the buyer and supplier idiosyncratic investment constructs share two items that measure each party's investments in both physical and human assets in the relationship, the third item for *buyer idiosyncratic investments* measures how much the buyer has trained the supplier to manufacture these products, and the third item for *supplier idiosyncratic investments* measures how much knowledge the supplier has tailored to the buyer's components. This distinction is consistent with the manufacturing outsourcing context in this study, where the buyer is outsourcing manufacturing and training the supplier to manufacture the outsourced components. Thus, increases in *buyer idiosyncratic investments* mean that the buyer has shared more information, making the buyer more susceptible to supplier poaching (i.e., increased opportunity to poach). *Product customization* is a single-item measure capturing the extent to which the supplier views the products as customized (1 = Industry standard components; 7 = Completely customized components) taken from Heide and Miner (1992). This measure was included because as products become more customized, the potential market for those products should decrease (i.e., reduce the opportunity to poach). It could be argued that some amount of customization would be required for poaching to be possible and then would fall off when products become highly customized (i.e., an inverse U-shaped relationship with poaching), which is an alternative we examine in the robustness checks section. A single-item measure, *new products,* captures the frequency that new products are introduced in the relationship (1 = Very infrequently; 7 = Very frequently). This measure was included because few new products introduced would mean little, if any, opportunity to poach. Finally, we also included a measure to control for the social desirability associated with poaching, which is outlined in section 4.3.

4.3. Common Method Bias and Social Desirability

There is some concern of common method bias because one of the independent variables, *supplier idiosyncratic investments*, comes from the survey that contains the dependent variable, *poaching*. Since poaching can be considered socially undesirable, this can lead to concerns of social desirability bias due to a common source (Podsakoff et al. 2003). To address this concern, we included an additional control variable, *poaching desirability*, which is intended to measure the item desirability of the poaching survey items. This measure was chosen based on the study by Randall and Fernandes (1991) examining methods for measuring social desirability, where they found that "perceived item desirability of the dependent variable appears to be preferable in future research" (p. 814). Additionally, we do not hypothesize a direct effect for *supplier idiosyncratic investments* and only hypothesize an interaction (H2). Interaction effects can only be attenuated from common method bias (Podsakoff et al. 2012, Siemsen et al. 2010), making any findings for H2 conservative. For the other three hypotheses, the independent and dependent variables come from different sources (i.e., the buyer survey or secondary sources), removing concerns of common method bias. Finally, we took additional steps to reduce the social desirability of poaching. We stressed the confidential and voluntary nature of the research project in all communications with respondents, which has been shown to increase the mean response levels on socially undesirable items (Berry et al. 2012). Additionally, we used an indirect questioning technique that has been shown to reduce the effects of social desirability (Fisher 1993).

4.4. Measurement Model

Using Stata 14.0, we first performed an exploratory factor analysis on all multi-item constructs because the media-rich communication construct was updated for this study. All items loaded significantly onto their intended constructs (loadings > 0.4), except for one media-rich communication item, which was subsequently dropped. Confirmatory factor analysis (CFA) was then performed on all multi-item constructs. Missing data were handled using full information maximum likelihood (FIML) (Allison 2002, Enders 2010). The CFA model exhibited sufficient overall model fit (RMSEA = 0.053; χ^2/df = 1.29; CFI = 0.959). All items loaded significantly on their intended constructs (all loadings > 0.4 with $p < 0.001$), exhibiting strong convergent validity. The model exhibited discriminant validity, as the average variance extracted (AVE) for each construct exceeded the square of all its inter-factor correlations (Fornell and Larcker 1981). Additionally, the covariance between constructs all differ significantly from 1 (Bagozzi et al. 1991). Table 2 details the descriptive statistics including composite reliability and AVE for the multi-item constructs.

5. Analysis

The primary research objectives were to investigate the effect of IP rights on supplier poaching and the effect of IP rights on the relationship between two transaction characteristics and poaching. Factor scores were used for all multi-item constructs. The correlation matrix is shown in Table 3. Right-skewed variables were natural log transformed before the analysis and are noted in Table 3. To improve the interpretability of the results and reduce multicollinearity, all multi-item constructs were standardized and secondary-measure independent variables were mean centered before the analysis (Cohen et al. 2003).

5.1. Endogeneity

Although we employ extensive control variables in our model, we still examine whether the coefficients of the independent variables could be biased due to omitted variables using an instrumental variable approach. Valid instruments must satisfy two conditions (Angrist and Pischke 2009, French and Popovici 2011, Wooldridge 2010): (1) valid instruments cannot be correlated with the error term in the second-stage equation and (2) should be highly significant in the

Table 2 Item Descriptive Statistics

Constructs and items	Mean	SD	Constructs and items	Mean	SD	Constructs and items	Mean	SD
Supplier idiosyncratic investments			Detailed contracts			Electronic information sharing		
(CR = 0.76; AVE = 0.52)			(CR = 0.92; AVE = 0.79)			(CR = 0.88; AVE = 0.61)		
SII1	4.68	1.79	CON1	4.55	1.83	EIS1	3.02	2.16
SII2	5.30	1.56	CON2	4.11	1.92	EIS2	5.17	1.92
SII3	4.75	1.64	CON3	3.95	2.00	EIS3	5.32	1.80
Media-rich communication			Poaching desirability			EIS4	5.22	1.99
(CR = 0.73; AVE = 0.58)			(CR = 0.98; AVE = 0.93)			EIS5	4.27	2.03
MRC1	3.85	1.53	PDES1	2.99	2.30	Buyer idiosyncratic investments		
MRC3	5.71	1.38	PDES2	3.01	2.25	(CR = 0.70; AVE = 0.45)		
Poaching (CR = 0.79; AVE = 0.56)			PDES3	3.14	2.28	BII1	4.15	1.88
PO1	1.55	1.19	Buyer size	3.00	2.06	BII2	3.25	1.70
PO2	1.73	1.45	ln(average distance)	4.96	3.93	BII3	3.43	1.80
PO3	1.53	1.08	New products	3.69	2.07	Relationship length	7.66	3.86
Patent	0.29	0.46	Sole source	1.97	0.86	Product customization	5.46	1.99
IP rights	7.69	1.15	Outsourcing degree	1.42	0.68	Supplier revenue	1.63	0.69

Notes: CR, Composite reliability; AVE, Average variance extracted.

Table 3 Correlation Matrix

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
1. ln(poaching)	1																
2. Poaching desirability	0.32	1															
3. Patent	−0.17	0.04	1														
4. Detailed contracts	−0.28	−0.06	0.19	1													
5. ln(average distance)	0.11	−0.14	−0.16	0.02	1												
6. Sole source	0.02	−0.10	−0.03	−0.05	0.06	1											
7. Outsourcing degree	−0.05	−0.11	0.17	0.16	−0.02	0.15	1										
8. Relationship length	−0.05	−0.05	0.09	−0.03	0.03	0.16	0.04	1									
9. Electronic info. sharing	−0.05	−0.09	0.11	0.25	0.08	0.13	0.16	0.09	1								
10. Buyer size	0.04	−0.09	0.00	0.26	0.06	0.05	0.24	0.12	0.12	1							
11. Supplier revenue	−0.03	−0.11	0.23	0.28	0.20	−0.07	0.32	0.21	0.31	0.14	1						
12. Buyer idiosyncratic inv.	0.13	0.03	0.17	0.64	0.00	0.02	0.29	−0.11	0.47	0.29	0.35	1					
13. Product customization	−0.18	0.00	0.08	0.16	−0.12	−0.20	−0.02	0.16	0.07	−0.01	0.10	0.06	1				
14. New products	0.01	−0.02	0.11	0.22	0.18	0.17	0.08	0.12	0.32	0.08	0.20	0.32	0.13	1			
15. Supplier idiosyncratic inv.	−0.10	0.06	−0.02	0.23	0.11	−0.03	0.09	0.09	0.42	0.21	0.38	0.25	0.20	0.32	1		
16. Media-rich comm.	−0.20	−0.23	0.06	0.37	−0.06	0.19	0.22	0.06	0.55	0.28	0.29	0.50	0.16	0.34	0.54	1	
17. IP rights	−0.20	0.03	−0.08	−0.12	−0.24	0.07	−0.10	0.07	−0.13	0.02	−0.34	−0.22	0.08	0.02	−0.09	0.03	1

Notes: Shaded regions depict highly significant values (two-sided $p < 0.05$).

first-stage equation once the effects of other exogenous variables (i.e., the covariates) have been netted out. To examine the former condition requires a model to have more instruments than potentially endogenous regressors (i.e., overidentified). With an overidentified model, we can perform tests of overidentifying restrictions, which would return a significant p-value if the instrumental variables are correlated with the error term. To examine the latter condition, we report the first-stage equations. When utilizing multiple instruments per potentially endogenous regressor, at least one needs to be highly significant (Angrist and Pischke 2009). We also discuss whether the F-statistic on the excluded instruments is above the commonly viewed threshold for strong instruments of 10 (French and Popovici 2011, Staiger and Stock 1997, Stock et al. 2002). After demonstrating the validity of our instruments, we perform the Durbin and Wu-Hausman tests of endogeneity, which test the null hypotheses that the variables are exogenous. We tested three separate 2SLS models, one for each potentially endogenous variable, where each model contained the covariates and one of the potentially endogenous variables.

To instrument *IP rights*, we used lagged measures of economic activity that have been employed as instruments for property rights in related research studies (e.g., Gould and Gruben 1996, Massimino et al. 2017): (1) GDP and (2) exports as a percentage of GDP. We lagged the instruments to reduce the correlation with the error term in the second-stage equation, utilizing observations from 1982, which was the earliest year that the World Bank (2015) had data available for all the supplier countries in our study. While secondary data were employed to instrument the secondary data measure of *IP rights*, variables from the primary data collection were used to instrument the other two independent variables. *Supplier idiosyncratic investments* were instrumented with a three-item measure of *supplier dependence* and a single-item measure of *supplier satisfaction*. *Supplier dependence* was used because previous research has found that firms invest more into idiosyncratic investments when higher percentages of a firm's output go to a single customer (Zaheer and Venkatraman 1995). *Supplier satisfaction* with the relationship was used as an instrument because previous research has demonstrated an association between a party's satisfaction in an exchange and that party's idiosyncratic investments (Ping 1993). *Media-rich communication* was instrumented with two single-item measures: (1) a measure of the supplier's *delivery frequency* and (2) a measure of the supplier's view of extending this relationship, *supplier extendedness*. *Delivery frequency* was utilized because more frequent deliveries should lead to increases in communication. *Supplier extendedness* was utilized because previous research has demonstrated an association between extendedness and relational governance activities, such as information exchange (Heide and Miner 1992). The survey questions used for the instruments can also be found in Appendix A.

Results for the first-stage equations are displayed in Table 4. Two of the models were run with a sample of 103 observations due to missing data for the instruments. All instruments were significant in their first-stage equations ($p < 0.05$), and at least one was highly significant in each equation ($p < 0.001$). Additionally, the lowest F-statistic on the excluded instruments was 13. The Sargan and Basmann tests of overidentifying restrictions were not significant for any of the models (p-values > 0.35). These results provide strong support for the use of all of our instruments. For *IP rights*, *supplier idiosyncratic investments*, and *media-rich communication*, the Durbin and Wu-Hausman tests were not able to reject the null hypotheses that the variables are exogenous (p-values > 0.99, 0.50, 0.25, respectively). Based on all of these results, the main analysis was run using hierarchical linear regression.

5.2. Econometric Model

The analysis was performed using hierarchical linear regression. Related blocks of variables were entered into the model sequentially to examine the explanatory power of each group. Stage 1 contains the control variables. In Stage 2, the transaction characteristics variables, *supplier idiosyncratic investments* and *media-rich communication*, are added to the model. *IP rights* are added in Stage 3, which enables an examination of the explanatory power of IP rights while controlling for the transaction characteristics. Both interaction terms are added in Stage 4 to examine whether the effects of the transaction characteristics are contingent on *IP rights*. The largest variance inflation factor across all stages was 2.82. Results of the analysis are presented in Table 5.

6. Results

Before discussing our research hypotheses, we make some observations regarding the control variables. First, the regression coefficient for *poaching desirability* is highly significant across all stages of the model, highlighting the importance of including a measure of social desirability in studies examining self-reports of opportunism (Podsakoff et al. 2003). Next, we included two measures of legal protections for poaching—*detailed contracts* and *patent*. The regression coefficient for *detailed contracts* is highly significant throughout all stages of the model, and the regression

Table 4 First-Stage Equations from Endogeneity Analysis

	Intellectual property rights		Sup. idiosyncratic investments		Media-rich communication	
	Coefficient	SE	Coefficient	SE	Coefficient	SE
Constant	−1.904***	0.281	−1.927***	0.608	−3.449***	0.673
Poaching desirability	−0.015	0.023	0.081**	0.037	−0.121***	0.041
Patent	−0.075	0.110	−0.301*	0.173	0.025	0.196
Detailed contracts	0.001	0.041	0.069	0.064	0.022	0.071
ln(average distance)	−0.020	0.013	−0.004	0.020	−0.032	0.023
Sole source	0.027	0.059	−0.013	0.093	0.164	0.104
Outsourcing degree	0.072	0.077	0.032	0.131	0.064	0.134
Relationship length	−0.002	0.013	−0.019	0.021	−0.026	0.023
Electronic information sharing	−0.071	0.055	0.253***	0.088	0.182*	0.103
Buyer size	0.020	0.025	0.065	0.040	0.013	0.044
Supplier revenue	0.044	0.087	0.187	0.145	0.035	0.146
Buyer idiosyncratic investments	−0.043	0.055	−0.123	0.086	0.204**	0.095
Product customization	−0.058**	0.026	0.054	0.040	0.031	0.044
New products	0.042*	0.025	0.070*	0.041	0.076*	0.044
Lagged GDP	0.772***	0.039				
Lagged % exports	0.012***	0.003				
Supplier dependence			0.534***	0.139		
Supplier satisfaction			0.185**	0.072		
Delivery frequency					0.339***	0.076
Supplier extendedness					0.201**	0.090
Number of observations	104		103		103	
R^2	0.858		0.512		0.622	
Adjusted R^2	0.834		0.428		0.557	
Root MSE	0.470		0.740		0.807	

Notes: $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table 5 Results

	1		2		3		4	
	Coefficient	SE	Coefficient	SE	Coefficient	SE	Coefficient	SE
Constant	−1.070**	0.611	−1.529**	0.592	−1.469**	0.587	−1.215**	0.572
Poaching desirability	0.353***	0.106	0.325***	0.113	0.334***	0.112	0.372***	0.109
Patent	−0.322	0.244	−0.387	0.245	−0.409*	0.243	−0.432*	0.235
Detailed contracts	−0.697***	0.138	−0.669***	0.138	−0.662***	0.137	−0.638***	0.132
ln(average distance)	0.042	0.029	0.032	0.029	0.022	0.029	0.013	0.029
Sole source	0.008	0.132	0.043	0.134	0.043	0.132	0.047	0.128
Outsourcing degree	−0.140	0.173	−0.142	0.171	−0.139	0.169	−0.045	0.168
Relationship length	0.010	0.030	0.006	0.030	0.011	0.029	0.014	0.028
Electronic info. sharing	−0.190	0.126	−0.085	0.136	−0.097	0.135	−0.155	0.132
Buyer size	0.035	0.055	0.057	0.056	0.065	0.055	0.030	0.055
Supplier revenue	0.050	0.186	0.115	0.193	0.031	0.197	−0.057	0.193
Buyer idiosyncratic inv.	0.720***	0.165	0.744***	0.169	0.693***	0.170	0.693***	0.164
Product customization	−0.064	0.056	−0.045	0.057	−0.040	0.056	−0.073	0.055
New products	−0.001	0.057	0.021	0.057	0.034	0.057	0.038	0.055
Supplier idiosyncratic inv.			−0.080	0.142	−0.094	0.141	0.038	0.144
Media-rich comm.			−0.231*	0.161	−0.189	0.161	−0.153	0.159
IP rights (IPR)					−0.174**	0.101	−0.154*	0.097
Sup. idiosyncratic inv. × IPR							−0.392***	0.134
Media-rich comm. × IPR							0.216**	0.109
F	4.32		4.06		4.08		4.37	
Prob > F	0.000		0.000		0.000		0.000	
F relative to prior	–		1.86		2.97		4.28	
Prob > F rel. to prior	–		0.162		0.088		0.017	
R^2	0.384		0.409		0.429		0.481	
Adjusted R^2	0.295		0.308		0.324		0.371	
Root MSE	1.051		1.041		1.030		0.993	

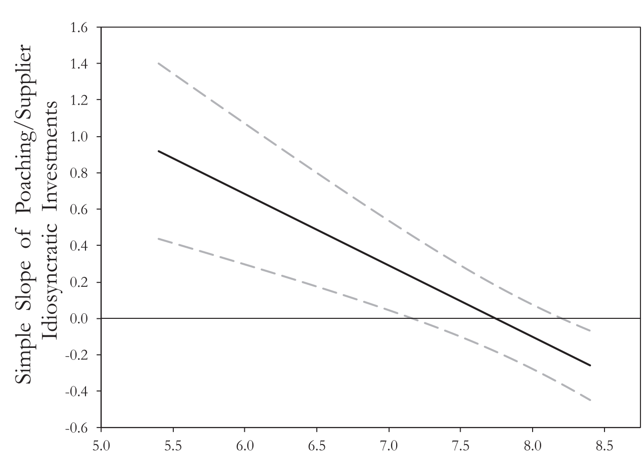
Notes: *p < 0.10; **p < 0.05; ***p < 0.01; One-tail t–test used when a directionally specific effect was posited, two-tail tests used otherwise.

coefficient for *patent* is significant in Stages 3 and 4 of the model. These findings illustrate the importance of controlling for legal instruments when examining IP risk. Finally, a significant positive relationship was found between *buyer idiosyncratic investments* and *poaching*, which illustrates that, consistent with the perspective of transaction cost economics, these investments open the buyer up to opportunism. No significant relationships were found for the other covariates.

Drawing upon arguments from deterrence theory, in H1, it was hypothesized that the strength of IP rights would have a significant negative effect on supplier poaching. In Stage 3 of the model, the coefficient for *IP rights* is in the expected direction and significant ($\beta = -0.174$, $p < 0.05$), and it is still significant in Stage 4, although at a reduced significance level ($p < 0.10$). Thus, H1 is supported.

For H2, it was hypothesized that *IP rights* negatively moderates the relationship between *supplier idiosyncratic investments* and *poaching*. In Stage 4, the coefficient for the interaction of *supplier idiosyncratic investments* and *IP rights* is in the expected direction and highly significant ($\beta = -0.392$, $p < 0.01$), confirming an interaction effect in the expected direction. Thus, H2 is strongly supported. While the significant interaction effect supports H2, we also want to examine the arguments that were made in support of H2. Specifically, we argued that *supplier idiosyncratic investments* would be negatively related to *poaching* when *IP rights* are strong and that these same investments would be positively related to *poaching* when *IP rights* are weak. To examine these arguments, we

Figure 2 Plot of the Conditional Effect of Supplier Idiosyncratic Investments on Poaching Across the Range of Intellectual Property Rights



Note. 90% confidence band depicted.

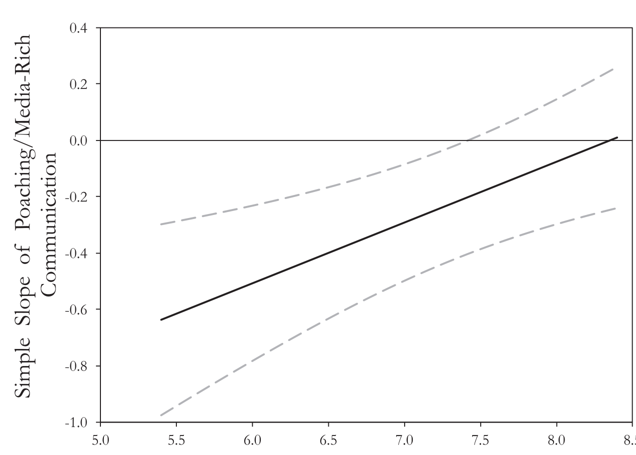
probe the interaction effect using the Johnson–Neyman technique[1] (Johnson and Neyman 1936). In Figure 2, we plot the conditional effect of *supplier idiosyncratic investments* on *poaching* (i.e., value of the simple slope) and the confidence band for that conditional effect over the observed range of the moderator, *IP rights,* with the other variables held at their mean level. Figure 2 illustrates the direction and magnitude of the effect that *supplier idiosyncratic investments* has on *poaching*. In Figure 2, positive (negative) values on the *y*-axis represent a positive (negative) relationship (i.e., the simple slope) between *supplier idiosyncratic investments* and *poaching* at specific values of *IP rights*, and that effect is significant when the confidence band does not contain zero. For reference, the 2014 *IP rights* scores for the United States and China are 8.4 and 5.4, respectively, and overlaying the *IP rights* scores against the strong/weak categorization in Zhao (2006), strong *IP rights* countries have *IP rights* values of 7.9 or higher. Figure 2 illustrates that the conditional effect of *supplier idiosyncratic investments* on *poaching* is negative for high values of *IP rights*. Specifically, there is a significant negative relationship between *supplier idiosyncratic investments* and *poaching* at the 10% level at *IP rights* values of 8.20 and higher (i.e., when the confidence band does not contain zero for high levels of the moderator). Additionally, although not graphically presented, this relationship becomes significant at the 5% level at an *IP rights* value of 8.35. Thus, consistent with the arguments used to develop H2, *supplier idiosyncratic investments* have a negative (i.e., safeguarding) effect on *poaching* when *IP rights* are strong. Conversely, that effect becomes positive for values of *IP rights* lower than 7.75. The argument that *supplier idiosyncratic investments* are positively related to *poaching* under low values of *IP rights* is also supported as the conditional effect is positive and significant at the 10% level at an *IP rights* value of 7.17. Additionally, although not graphically presented, this effect becomes significant at the 5% level at an *IP rights* value of 6.89. These findings provide strong support for the argument that *supplier idiosyncratic investments* act as a safeguard to supplier *poaching* when *IP rights* are strong, but these investments are associated with an increase in *poaching* when *IP rights* are weak.

For H3, it was hypothesized that *media-rich communication* between the buyer and supplier would be negatively associated with *poaching* independent of the strength of *IP rights*. *Media-rich communication* was added in Stage 2 of the model and was modestly significant ($p < 0.10$), but it was not significant in Stages 3 or 4. Thus, H3 is not supported. For H4, it was hypothesized that the effect of *media-rich communication* on *poaching* would be stronger when *IP rights* are weak. In Stage 4, the regression coefficient associated with the *media-rich communication* and *IP rights* interaction term is in the predicted direction and significant ($\beta = 0.216$, $p < 0.05$), confirming that *media-rich communication* has a stronger safeguarding (negative) effect when *IP rights* are weak. Thus, H4 is supported. The interaction is graphically depicted in Figure 3, which illustrates that as *IP rights* become weaker the safeguarding effect of *media-rich communication* becomes stronger. Specifically, Figure 3 shows that the relationship becomes significant at the 10% level at an *IP rights* value of 7.42. Additionally, although not graphically presented, this relationship becomes significant at the 5% level at an *IP rights* value of 7.14. Overall, Figure 3 illustrates that, when *IP rights* are weak, *media-rich communication* is negatively associated with *poaching*, but, when *IP rights* are strong, there is no relationship between *media-rich communication* and *poaching*. This results in the average effect of *media-rich communication* (i.e., the effect posited in H3) not differing from zero.

6.1. Robustness Checks

We ran several alternative models to examine the robustness of the specification of our model. First, the analysis was also run using minimum and maximum distance, and the coefficients of the variables of interest did not significantly differ with either of these specifications. Next, we examined substituting the overall IPRI index in place of the IP rights measure, and the results are consistent across both specifications. We also examined a model that contained a quadratic *product customization* term because it could be argued that some amount of customization would be required for poaching to be possible but that poaching would fall off when products become highly customized. The results are consistent with

Figure 3 Plot of the Conditional Effect of Media-Rich Communication on Poaching Across the Range of Intellectual Property Rights



Note. 90% confidence band depicted.

this term included. Additionally, we examined whether there was an interaction effect between the *IP rights* and *electronic information sharing* that could substitute for the *media-rich communication* and *IP rights* interaction. To do this, we tested two models. The first model added an interaction term between *IP rights* and *electronic information sharing* to the full model, and the second model removed the interaction between *media-rich communication* and *IP rights* but kept the *IP rights* and *electronic information sharing* interaction. Including the *IP rights* and *electronic information sharing* interaction in the main model did not affect the results of the analysis, and the *IP rights* and *electronic information sharing* term was not significant in either model. Additionally, we examined whether either mode of media-rich communication used in the multi-item construct (i.e., face-to-face or telephone) had a stronger effect than the other. To examine different modes of media-rich communication, we ran two separate models each using the single-item indicator. The results of the hypotheses did not differ across these models, supporting the use of the multi-item media-rich communication construct. Finally, we examined whether the effects of two societal cultural values that have been related to increases in loyalty and commitment, collectivism and future orientation, could affect the observed relationships. The cultural values were drawn from the GLOBE study (House et al. 2004). We ran two additional models, with each model adding a cultural value to the model in Stage 4 of the main analysis. The coefficients for the variables supporting our hypotheses in these additional models were not significantly different than those in the main analysis.

7. Conclusion

7.1. Theoretical Implications

The finding that the strength of IP rights moderates the relationship between a supplier's investments in specialized assets and the supplier engaging in a specific form of opportunism, poaching, adds to the transaction cost economics (TCE) literature. In the buyer–supplier literature drawing from TCE, a supplier's idiosyncratic investments have been considered a safeguard for buyers from supplier opportunism (e.g., Achrol and Gundlach 1999, Gundlach et al. 1995, Jap and Anderson 2003, Katsikeas et al. 2009, Liu et al. 2009, Wathne and Heide 2000, Williamson 1985). Conversely, others (e.g., Aron et al. 2005, Teece 1986) have argued that a supplier's investments in specialized assets can lead to increases in supplier poaching. Our findings resolve these contradictory arguments by illustrating that, depending on the strength of the IP rights where the supplier is located, a supplier's investments in specialized assets could either act as a safeguard from poaching or lead to increases in poaching. When idiosyncratic investments by a supplier lead to a reduction in supplier opportunism, it is because these assets have substantially lower value outside of that transaction, or in other words are transaction or relationship specific assets. The specialized nature of these investments creates a small numbers bargaining situation (i.e., few possible exchange partners) and an *ex-post* dependency in the relationship because, if that relationship is terminated, the firm will forgo the value of the idiosyncratic investment. Changes in the strength of IP rights can alter this small numbers bargaining situation. Under the condition of weak IP rights, not only did supplier idiosyncratic investments not act as a safeguarding mechanism (i.e., associated with a reduction in poaching), but these investments also were found to be associated with an increase in poaching. Thus, under this condition, these investments, while specialized to a product, are not necessarily transaction or relationship specific because the products that these assets produce can be sold elsewhere through supplier poaching (i.e., weak IP rights can increase the number of possible exchange partners). In sum, these findings illustrate that the strength of IP rights is one of the determinants of the degree to which idiosyncratic investments are transaction specific. Put simply, strong IP rights are a boundary condition for the safeguarding effect of investments in specialized assets. The current study has clear implications for TCE as it demonstrates that the strength of IP rights should be an explicit consideration in studies examining idiosyncratic investments in different institutional environments, as investments that are transaction specific in one location might not be transaction specific in another.

The finding that, when managing suppliers in weak IP rights locations, media-rich communication from a manufacturer's boundary spanners can safeguard the manufacturer from supplier poaching has implications for multiple literature streams. First, this study adds to the relational governance literature that has examined how to protect buyer–supplier relationships in emerging markets from opportunism (e.g., Liu et al. 2009, Luo 2007, Peng 2003, Zhou and Xu 2012) by showing that relational governance can also be a safeguard for an additional form of opportunism, poaching. Next, this study adds to the emerging literature on protecting a firm's IP in weak IP rights locations (e.g., Holmes et al. 2016, Keupp et al. 2009, 2010, Massimino et al. 2017). While the existing body of literature has shown that building relationships can be effective in protecting IP for MNEs operating in weak property rights locations, this study is the first to illustrate that relational governance can also be effective at protecting IP for manufacturers outsourcing to suppliers in these locations. Finally, this study adds to the literature on boundary spanning (e.g., Anderson et al. 2017, Levina and Vaast 2005, 2008, Orlikowski 2002, Parker and Anderson 2002). Research on boundary spanning (e.g., Levina and Vaast 2008, Orlikowski 2002) has illustrated how media rich modes of communication can overcome multiple boundaries, such as inter-organizational, cultural, time zones, and geographic distance. This study adds to that literature by illustrating how inter-organizational boundary spanners are also able to overcome differences created by engaging with supply chain partners in countries with weak IP rights. The findings in this study further illustrate an additional soft skill of supply chain integrators (Parker and Anderson 2002), which is being able to create a relationship between the manufacturer and a supplier that can act as a safeguard from opportunism.

Overall, this study adds to the literature examining how the institutional environment influences the effectiveness of different governance mechanisms and to the literature that examines how to profit from innovation. This study adds to the literature utilizing Williamson's (1991) shift parameter framework (e.g., Handley and Angst 2015, Oxley 1999), which posits that the effectiveness of different governance mechanisms is contingent on the institutional environment, by illustrating how two different types of safeguards are contingent on the strength of IP rights. This study also adds to the literature utilizing Teece's (1986) profiting from innovation (PFI) framework, which posits that in the absence of strong IP rights a firm can protect its IP by combining it with a complementary asset or resource. This study demonstrates that relationships with suppliers, which are considered a complementary resource (Helfat and Lieberman 2002), can be utilized as a method to protect a manufacturer's IP, which is consistent with the PFI framework.

7.2. Managerial Implications

The findings in this study illustrate that a one-size-fits-all supplier management strategy will not work to control poaching across suppliers in locations with varying levels of IP rights. In the absence of a strong legal system, managers must find alternative mechanisms, such as safeguards, to protect their IP. Thus, understanding which safeguards will be the most effective is of critical practical importance. A supplier's investments into specialized assets have previously been considered a safeguard against supplier opportunism because they can create a small numbers bargaining situation for the supplier. This study illustrates that these investments can be associated with an increased risk of poaching for suppliers in countries with weak IP rights because changes in the strength of IP rights can affect that small numbers bargaining situation. These investments could lead manufacturers to have a false sense of security in regards to their IP, as the supplier may be investing in the relationship to learn as much as it can from the buyer to redeploy those investments outside of the relationship with the buyer. Thus, when outsourcing to suppliers in locations with weak IP rights, manufacturers should consider whether or not a supplier's investments are truly transaction specific because weak IP rights can affect the extent to which these investments are redeployable. This is not to say that there are no instances in weak IP rights locations where a supplier's idiosyncratic investments can lead to a small numbers bargaining situation. There are situations where possible exchange partners outside of the current buyer are few, such as modularized components that need to be combined with other complex technology to be functional (Gassmann et al. 2012). What this study illustrates is that manufacturers need to explicitly consider if the legal system is the mechanism that would make these investments transaction specific prior to outsourcing to a supplier in a country with weak IP rights.

When changes in IP rights alter the small numbers bargaining situation that can be created by suppliers' idiosyncratic investments, manufacturers must look to other safeguards. Specifically, we show that the relational governance formed by engaging in media-rich communication can be that safeguard. Practically, this highlights the importance of boundary-spanning personnel, especially the amount of time that supply chain professionals spend on late-night phone calls with suppliers in different time zones and long trips overseas building relationships with suppliers. For manufacturers engaged in offshore outsourcing, our findings illustrate the importance of supply chain personnel that can foster relationships. Unfortunately, as Parker and Anderson (2002) have noted, some firms have eliminated the managers that were often crucial in spanning the boundaries created by offshore outsourcing. Additionally, as Levina and Vaast (2008) discovered, even when boundary-spanning personnel are in place, their ability to engage in media rich modes of communication can often be financially constrained by travel budgets. Indeed, discussions with one of the participants in this research study highlighted similar constraints, as the respondent discussed recurring pressure to reduce travel budgets and communicate more electronically. This study cautions against that mentality and echoes a sentiment similar to Frazier (1999), who cautioned firms not to focus solely on electronic information sharing and ignore the importance of boundary-spanning personnel. Overall, our findings can assist manufacturers in managing IP risk, which is one of the main hidden costs of outsourcing (Burton 2013).

7.3. Limitations and Future Research

Like all research studies, this study has several limitations, some of which can be filled by future research. Future research could examine whether the findings from this study extend to other types of outsourcing, for example business process outsourcing. The cross-sectional nature of our data limits assertions of causality. A longitudinal study with a larger sample would be an ideal direction for future research. Another possible extension of this study would be to examine different types of investments in specialized assets. There are six different types of specialized assets: physical, human, site, brand name capital, temporal, and dedicated assets (Williamson 1996). This study only investigated two types of idiosyncratic investments: physical and human. Although site specificity, where "successive stages of production are located in a cheek-by-jowl relation" (Williamson 1996, p. 59), may not be possible in offshore outsourcing relationships, future research could examine if the remaining three types of idiosyncratic investments have relationships similar to what was found in this study. Additionally, a key assumption in our theorizing is that media-rich communication is fundamentally a relational construct. While this assumption is consistent with the literature discussing that media-rich modes of communication are instrumental in building a relationship (e.g., He and Nickerson 2006, Uzzi 1997), we acknowledge that we do not directly observe relational governance in the dyads in our study and we can only infer that relational governance results from media-rich communication. Thus, another area for future research could be to further investigate the safeguarding relationship we observe from media-rich communication. For example, using case studies to perform a more fine-grained examination of how these modes of communication affect the relationship between a buyer and supplier could yield additional insights into managing outsourcing relationships. Future research could also examine if other facets of relational governance, such as trust, have similar effectiveness in controlling poaching. In this study, we focused on poaching, however, understanding how the institutional environment affects other forms of opportunism is also an interesting area for future research, as understanding how the institutional environment can affect other hidden costs of outsourcing is both practically and theoretically relevant.

Acknowledgments

The authors thank the departmental editor Edward Anderson, the senior editor, and the three anonymous reviewers for their constructive feedback on the article. This research was partially supported by a grant from the National Center for the Middle Market at The Ohio State University's Fisher College of Business.

Appendix A: Survey Questions

Supplier Idiosyncratic Investments (Source = Supplier) (1 = Strongly disagree; 7 = Strongly agree).

SII1 If the relationship with <<BUYER>> was terminated, we would be wasting a lot of knowledge that's tailored to their components.

SII2 We have made a substantial investment in personnel dedicated to <<BUYER>>.

SII3 We have made a substantial investment in capital equipment and technology dedicated to <<BUYER>>

Media-Rich Communication (Source = Buyer) (1 = Never; 2 = Annually; 3 = Semi-annually; 4 = Quarterly; 5 = Monthly; 6 = Weekly; 7 = Daily)

How often do personnel from your firm and <<SUPPLIER>> communicate using the following methods. . .

PC1 Face-to-face

PC2 Video conferencing**

PC3 Telephone

**Item dropped due to EFA results

Poaching (Source = Supplier) (1 = Very unlikely; 7 = Very likely)

In dealings with <<BUYER>>, how likely are members of your organization to use potentially proprietary information obtained through your relationship with <<BUYER>>. . .

PO1 to gain favor with other clients.

PO2 to help win new business with other customers.

PO3 to develop new products that you can offer in the marketplace

Detailed Contracts (Source = Buyer) (1 = Strongly disagree; 7 = Strongly agree)
CON1 We have specific, well-detailed agreements with <<SUPPLIER>>.
CON2 We have customized agreements that detail the obligations of both parties.
CON3 We have detailed contractual agreements specifically designed with <<SUPPLIER>>

Electronic Information Sharing (Source = Buyer) (1 = Never; 2 = Annually; 3 = Semi-annually; 4 = Quarterly; 5 = Monthly; 6 = Weekly; 7 = Daily)
How often does <<SUPPLIER>> electronically provide information about. . .
ES1 Production capacity information
ES2 Order status information
ES3 Delivery schedule information
ES4 Changes in delivery schedule
ES5 Lead time information for products

Buyer Idiosyncratic Investments (Source = Buyer) (1 = Strongly disagree; 7 = Strongly agree)
BII1 We have made a substantial investment in personnel dedicated to <<SUPPLIER>>.
BII2 We have made a substantial investment in capital equipment and technology dedicated to <<SUPPLIER>>.
BII3 We give extensive training to <<SUPPLIER>> on how to manufacture these components

Poaching Desirability (Source = Supplier) (1 = Very uneasy; 7 = Not at all uneasy)
Survey questions sometimes have different kinds of effects on people. We would like your opinion about some of the questions in this survey. For each of the questions below, please indicate on the provided scale the extent to which you think those questions would make most people uneasy.
PDES1 In dealings with <<BUYER>>, how likely are members of your organization to use potentially proprietary information obtained through your relationship with <<BUYER>> to gain favor with other clients?
PDES2 In dealings with <<BUYER>>, how likely are members of your organization to use potentially proprietary information obtained through your relationship with <<BUYER>> to help win new business with other customers?
PDES3 In dealings with <<BUYER>>, how likely are members of your organization to use potentially proprietary information obtained through your relationship with <<BUYER>> to develop new products that you can offer in the marketplace?

Supplier Dependence (Source = Supplier) (1 = Strongly disagree; 7 = Strongly agree)
SDEP1 If the relationship with <<BUYER>> was terminated, it would not hurt our operations. *Reverse coded
(1 = < 1%; 2 = 1–4%; 3 = 5–20%; 4 = 21–35%; 5 = 36–50%; 6 = 51–65%; 7 = 66–80%; 8 = > 80%)
SDEP2 What percentage of your organization's total revenue does <<BUYER>> account for?
SDEP3 What percentage of your organization's total profits does <<BUYER>> account for?

Supplier Satisfaction (Source = Supplier) (1 = Strongly disagree; 7 = Strongly agree)
The relationship with <<BUYER>> has greatly benefited our organization.

Delivery Frequency (Source = Buyer) (1 = Annually; 2 = Semi-annually; 3 = Quarterly; 4 = Monthly; 5 = Weekly; 6 = 2–3 times per week; 7 = Daily)
On average, how often do you receive deliveries from <<SUPPLIER>>?

Supplier Extendedness (Source = Supplier) (1 = Strongly disagree to 7 = Strongly agree)
It is assumed that renewal of agreements in this relationship will generally occur.

Note

[1]We utilize this technique, as opposed to that proposed by Aiken and West (1991), because the choice of high and low values when using that method can be considered arbitrary (Preacher et al. 2006).

References

Achrol, R. S., G. T. Gundlach. 1999. Legal and social safeguards against opportunism in exchange. *J. Retail*. **75**(1): 107–124.

Aiken, L. S., S. G. West. 1991. *Multiple Regression: Testing and Interpreting Interactions*. Sage Publications, Thousand Oaks, CA.

Alchian, A. A., H. Demsetz. 1972. Production, information costs, and economic organization. *Am. Econ. Rev.* **62**(5): 777–795.

Allison, P. D. 2002. *Missing Data*. Sage Publications, Thousand Oaks, CA.

Anand, B., A. Galetovic. 2004. How market smarts can protect property rights. *Harv. Bus. Rev.* **82**(12): 72–79.

Anderson, E., B. Weitz. 1989. Determinants of continuity in conventional industrial channel dyads. *Market. Sci.* **8**(4): 310–323.

Anderson, E., B. Weitz. 1992. The use of pledges to build and sustain commitment in distribution channels. *J. Market. Res.* **29**(1): 18–34.

Anderson, E. G., A. Chandrasekaran, A. Davis-Blake, G. G. Parker. 2017. Managing distributed product development projects: Integration strategies for time zone and language barriers. *Information Systems Research*. Forthcoming. https://doi.org/10.1287/isre.2017.0733

Angrist, J. D., J. Pischke. 2009. *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press, Princeton.

Aral, S., Y. Bakos, E. Brynjolfsson. 2017. Information technology, repeated contracts, and the number of suppliers. *Management Sci.* Articles in Advance, pp. 1–21. https://doi.org/10.1287/mnsc.2016.2631.

Aron, R., E. K. Clemons, S. Reddi. 2005. Just right outsourcing: Understanding and managing risk. *J. Manag. Inf. Syst.* **22**(2): 37–55.

Bagozzi, R. P., Y. Yi, L. W. Phillips. 1991. Assessing construct validity in organizational research. *Adm. Sci. Q.* **36**(3): 421–458.

Berry, C. M., N. C. Carpenter, C. L. Barratt. 2012. Do other-reports of counterproductive work behavior provide an incremental contribution over self-reports? A meta-analytic comparison. *J. Appl. Psychol.* **97**(3): 613–636.

Brown, J. R., C. S. Dev, D. Lee. 2000. Managing marketing channel opportunism: The efficacy of alternative governance mechanisms. *J. Market.* **64**(2): 51–65.

Burton, T. T. 2013. The 10 hidden costs of outsourcing. *CSCMP's Supply Chain Q.* (Quarter 2/2013): 30–35.

Carlson, J. R., J. F. George. 2004. Media appropriateness in the conduct and discovery of deceptive communication: The relative influence of richness and synchronicity. *Group Decis. Negot.* **13**(2): 191–210.

Chesbrough, H., S. Ahern, M. Finn, S. Guerraz. 2006. Business models for technology in the developing world: The role of non-governmental organizations. *Ca. Manag. Rev.* **48**(3): 48–61.

Cho, S., X. Fang, S. Tayur. 2015. Combating strategic counterfeiters in licit and illicit supply chains. *Manuf. Serv. Oper. Manag.* **17**(3): 273–289.

Cohen, J., P. Cohen, S. G. West, L. S. Aiken. 2003. *Applied Multiple Regression/Correlation Analysis for the Behavioral Sciences*, 3rd edn. L. Erlbaum Associates, London, United Kingdom.

Commission on the Theft of American Intellectual Property. 2013. *The IP Commission report: The report of the Commission on the Theft of American Intellectual Property*. Report, The National Bureau of Asian Research, United States of America.

Daft, R. L., R. H. Lengel. 1986. Organizational information requirements, media richness and structural design. *Management Sci.* **32**(5): 554–571.

Dillman, D. A., J. D. Smyth, L. M. Christian. 2009. *Internet, Mail, and Mixed-Mode Surveys: The Tailored Design Method*, 3rd edn. John Wiley & Sons, Hoboken, NJ.

Dwyer, F. R., P. H. Schurr, S. Oh. 1987. Developing buyer–seller relationships. *J. Market.* **51**(2): 11–27.

Dyer, J. H. 1996. Specialized supplier networks as a source of competitive advantage: Evidence from the auto industry. *Strateg. Manag. J.* **17**(4): 271–291.

Dyer, J. H., H. Singh. 1998. The relational view: Cooperative strategy and sources of interorganizational competitive advantage. *Acad. Manag. Rev.* **23**(4): 660–679.

Enders, C. K. 2010. *Applied Missing Data Analysis*. Guilford Press, New York.

Fisher, R. J. 1993. Social desirability bias and the validity of indirect questioning. *J. Consum. Res.* **20**(2): 303–315.

Fornell, C., D. F. Larcker. 1981. Evaluating structural equation models with unobservable variables and measurement error. *J. Market. Res.* **18**(1): 39–50.

Forza, C. 2002. Survey research in operations management: A process-based perspective. *Int. J. Oper. Prod. Manag.* **22**(2): 152–194.

Frazier, G. L. 1999. Organizing and managing channels of distribution. *J. Acad. Market. Sci.* **27**(2): 226–240.

French, M. T., I. Popovici. 2011. That instrument is lousy! In search of agreement when using instrumental variables estimation in substance use research. *Health Econ.* **20**(2): 127–146.

Gao, S. Y., W. S. Lim, C. S. Tang. 2017. Entry of copycats of luxury brands. *Market. Sci.* **36**(2): 272–289.

Gassmann, O., A. Beckenbauer, S. Friesike. 2012. *Profiting from Innovation in China*. Springer-Verlag, Berlin.

Givon, M., V. Mahajan, E. Muller. 1995. Software piracy: Estimation of lost sales and the impact on software diffusion. *J. Market.* **59**(1): 29–37.

Gould, D. M., W. C. Gruben. 1996. The role of intellectual property rights in economic growth. *J. Dev. Econ.* **48**(2): 323–350.

Granovetter, M. 1985. Economic action and social structure: The problem of embeddedness. *Am. J. Sociol.* **91**(3): 481–510.

Gray, J. V., G. Esenduran, M. J. Rungtusanatham, K. Skowronski. 2017. Why in the world did they reshore? Examining small to medium-sized manufacturer decisions. *J. Oper. Manag.* **49–51**: 37–51.

Gundlach, G. T., R. S. Achrol, J. T. Mentzer. 1995. The structure of commitment in exchange. *J. Market.* **59**(1): 78–92.

Handley, S. M., C. M. Angst. 2015. The impact of culture on the relationship between governance and opportunism in outsourcing relationships. *Strateg. Manag. J.* **36**(9): 1412–1434.

Handley, S. M., W. C. Benton. 2012. The influence of exchange hazards and power on opportunism in outsourcing relationships. *J. Oper. Manag.* **30**(1–2): 55–68.

He, D., J. A. Nickerson. 2006. Why do firms make and buy? Efficiency, appropriability and competition in the trucking industry. *Strateg. Organ.* **4**(1): 43–69.

Heide, J. B., A. S. Miner. 1992. The shadow of the future: Effects of anticipated interaction and frequency of contact on buyer–seller cooperation. *Acad. Manag. J.* **35**(2): 265–291.

Helfat, C. E., M. B. Lieberman. 2002. The birth of capabilities: Market entry and the importance of pre–history. *Ind. Corp. Change* **11**(4): 725–760.

Holcomb, T. R., M. A. Hitt. 2007. Toward a model of strategic outsourcing. *J. Oper. Manag.* **25**(2): 464–481.

Holmes, R. M., S. A. Zahra, R. E. Hoskisson, K. DeGhetto, T. Sutton. 2016. Two-way streets: The role of institutions and technology policy in firms' corporate entrepreneurship and political strategies. *Acad. Manag. Perspect.* **30**(3): 247–272.

House, R. J., P. J. Hanges, M. Javidan, P. W. Dorfman, V. Gupta. 2004. *Culture, Leadership, and Organizations: The GLOBE Study of 62 Societies*. Sage Publications, Thousand Oaks, CA.

IPRI. 2014. *International Property Rights Index 2014 Report*. Report, Property Rights Alliance, Washington, DC.

Jap, S. D., E. Anderson. 2003. Safeguarding interorganizational performance and continuity under ex post opportunism. *Management Sci.* **49**(12): 1684–1701.

Johnson, P. O., J. Neyman. 1936. Tests of certain linear hypotheses and their applications to some educational problems. *Stat. Res. Mem.* **1**: 57–93.

Katsikeas, C. S., D. Skarmeas, D. C. Bello. 2009. Developing successful trust-based international exchange relationships. *J. Int. Bus. Stud.* **40**(1): 132–155.

Keupp, M. M., A. Beckenbauer, O. Gassmann. 2009. How managers protect intellectual property rights in China using de facto strategies. *R&D Manag.* **39**(2): 211–224.

Keupp, M. M., A. Beckenbauer, O. Gassmann. 2010. Enforcing intellectual property rights in weak appropriability regimes: The case of de facto protection strategies in China. *Manag. Int. Rev.* **50**(1): 109–130.

Klein, B., R. G. Crawford, A. A. Alchian. 1978. Vertical integration, appropriable rents, and the competitive contracting process. *J. Law Econ.* **21**(2): 297–326.

Lawson, B., D. Samson, S. Roden. 2012. Appropriating the value from innovation: Inimitability and the effectiveness of isolating mechanisms. *R&D Manag.* **42**(5): 420–434.

Levina, N., E. Vaast. 2005. The emergence of boundary spanning competence in practice: Implications for implementation and use of information systems. *MIS Q.* **29**(2): 335–363.

Levina, N., E. Vaast. 2008. Innovating or doing as told? Status differences and overlapping boundaries in offshore collaboration. *MIS Q.* **32**(2): 307–332.

Liu, Y., Y. Luo, T. Liu. 2009. Governing buyer–supplier relationships through transactional and relational mechanisms: Evidence from China. *J. Oper. Manag.* **27**(4): 294–309.

Luo, Y. 2007. An integrated anti-opportunism system in international exchange. *J. Int. Bus. Stud.* **38**(6): 855–877.

Macneil, I. R. 1978. Contracts: Adjustment of long-term economic relations under classical, neoclassical and relational contract law. *Northwest. Univ. Law Rev.* **72**: 854–905.

Macneil, I. R. 1985. Relational contract: What we do and do not know. *Wis. Law Rev.* **1985**(3): 483–525.

Massimino, B., J. V. Gray, K. K. Boyer. 2017. The effects of agglomeration and national property rights on digital confidentiality performance. *Prod. Oper. Manag.* **26**(1): 162–179.

McIvor, R. 2009. How the transaction cost and resource-based theories of the firm inform outsourcing evaluation. *J. Oper. Manag.* **27**(1): 45–63.

Midler, P. 2009. *Poorly made in China: An Insider's Account of the Tactics Behind China's Production Game*. John Wiley & Sons, Hoboken, NJ.

Orlikowski, W. J. 2002. Knowing in practice: Enacting a collective capability in distributed organizing. *Organ. Sci.* **13**(3): 249–273.

Oxley, J. E. 1999. Institutional environment and the mechanisms of governance: The impact of intellectual property protection on the structure of inter-firm alliances. *J. Econ. Behav. Organ.* **38**(3): 283–309.

Parker, G. G., E. G. Anderson. 2002. From buyer to integrator: The transformation of the supply-chain manager in the vertically disintegrating firm. *Prod. Oper. Manag.* **11**(1): 75–91.

Paternoster, R. 2010. How much do we really know about criminal deterrence? *J. Crim. Law Criminol.* **100**(3): 765–823.

Peng, M. W. 2003. Institutional transitions and strategic choices. *Acad. Manag. Rev.* **28**(2): 275–296.

Ping, R. A. 1993. The effects of satisfaction and structural constraints on retailer exiting, voice, loyalty, opportunism, and neglect. *J. Retail.* **69**(3): 320–352.

Podsakoff, P. M., S. B. MacKenzie, J. Lee, N. P. Podsakoff. 2003. Common method biases in behavioral research: A critical review of the literature and recommended remedies. *J. Appl. Psychol.* **88**(5): 879–903.

Podsakoff, P. M., S. B. MacKenzie, N. P. Podsakoff. 2012. Sources of method bias in social science research and recommendations on how to control it. *Annu. Rev. Psychol.* **63**(1): 539–569.

Poppo, L., T. Zenger. 2002. Do formal contracts and relational governance function as substitutes or complements? *Strateg. Manag. J.* **23**(8): 707–725.

Porter, M. E. 1985. *Competitive Advantage: Creating and Sustaining Superior Performance*. Free Press, New York, NY.

Prahinski, C., W. C. Benton. 2004. Supplier evaluations: Communication strategies to improve supplier performance. *J. Oper. Manag.* **22**(1): 39–62.

Preacher, K. J., P. J. Curran, D. J. Bauer. 2006. Computational tools for probing interactions in multiple linear regression, multilevel modeling, and latent curve analysis. *J. Educ. Behav. Stat.* **31**(3): 437–448.

Qian, Y. 2014. Counterfeiters: Foes or friends? How counterfeits affect sales by product quality tier. *Management Sci.* **60**(10): 2381–2400.

Qian, Y., H. Xie. 2014. Which brand purchasers are lost to counterfeiters? An application of new data fusion approaches. *Market. Sci.* **33**(3): 437–448.

Quan, X., H. Chesbrough. 2010. Hierarchical segmentation of R&D process and intellectual property protection: Evidence from multinational R&D laboratories in China. *IEEE Trans. Eng. Manag.* **57**(1): 9–21.

Randall, D. M., M. F. Fernandes. 1991. The social desirability response bias in ethics research. *J. Bus. Ethics* **10**(11): 805–817.

Rogelberg, S. G., J. M. Stanton. 2007. Introduction: Understanding and dealing with organizational survey nonresponse. *Organ. Res. Methods* **10**(2): 195–209.

Roh, J. A., J. M. Whipple, K. K. Boyer. 2013. The effect of single rater bias in multi-stakeholder research: A methodological evaluation of buyer–supplier relationships. *Prod. Oper. Manag.* **22**(3): 711–725.

Rossetti, C., T. Y. Choi. 2005. On the dark side of strategic sourcing: Experiences from the aerospace industry. *Acad. Manag. Executive* **19**(1): 46–60.

Schotter, A., M. Teagarden. 2014. Protecting intellectual property in China. *Sloan Manag. Rev.* **55**(4): 40–48.

Shengjun, L., J. A. Fernandez. 2007. General Motors China versus Chery: Disputes over intellectual property rights. *Asian Case Res. J.* **11**(2): 269–292.

Siemsen, E., A. Roth, P. Oliveira. 2010. Common method bias in regression models with linear, quadratic, and interaction effects. *Organ. Res. Methods* **13**(3): 456–476.

Staiger, D., J. H. Stock. 1997. Instrumental variables regression with weak instruments. *Econometrica* **65**(3): 557–586.

Stock, J. H., J. H. Wright, M. Yogo. 2002. A survey of weak instruments and weak identification in generalized method of moments. *J. Bus. Econ. Stat.* **20**(4): 518–529.

Tangpong, C., K. Hung, Y. K. Ro. 2010. The interaction effect of relational norms and agent cooperativeness on opportunism in buyer–supplier relationships. *J. Oper. Manag.* **28**(5): 398–414.

Teece, D. J. 1986. Profiting from technological innovation: Implications for integration, collaboration, licensing and public policy. *Res. Policy* **15**(6): 285–305.

USITC. 2010. China: Intellectual property infringement, indigenous innovation policies, and frameworks for measuring the effects on the U.S. economy. Report, United States International Trade Commission, Washington, DC, USITC Publication 4199.

USITC. 2011. China: Effects of intellectual property infringement and indigenous innovation policies on the U.S. economy.

Report, United States International Trade Commission, Washington, DC. USITC Publication 4226.

Uzzi, B. 1997. Social structure and competition in interfirm networks: The paradox of embeddedness. *Adm. Sci. Q.* **42**(1): 35–67.

Wang, Q., J. J. Li, W. T. Ross, C. W. Craighead. 2013. The interplay of drivers and deterrents of opportunism in buyer–supplier relationships. *J. Acad. Market. Sci.* **41**(1): 111–131.

Wathne, K. H., J. B. Heide. 2000. Opportunism in interfirm relationships: Forms, outcomes, and solutions. *J. Market*. **64**(4): 36–51.

Williamson, O. E. 1979. Transaction-cost economics: The governance of contractual relations. *J. Law Econ.* **22**(2): 233–261.

Williamson, O. E. 1983. Credible commitments: Using hostages to support exchanges. *Am. Econ. Rev.* **73**(4): 519–540.

Williamson, O. E. 1985. *The Economic Institutions of Capitalism: Firms, Markets, Relational Contracting*. The Free Press, New York, NY.

Williamson, O. E. 1991. Comparative economic organization: The analysis of discrete structural alternatives. *Adm. Sci. Q.* **36**(2): 269–296.

Williamson, O. E. 1996. *The Mechanisms of Governance*. Oxford University Press, New York.

Wooldridge, J. M. 2010. *Econometric Analysis of Cross Section and Panel Data*, 2nd edn. MIT Press, Cambridge, Massachusetts.

World Bank. 2015. World Bank Indicators. Available at http://data.worldbank.org/indicator (accessed date June 24, 2015).

Yang, D., M. Sonmez, D. Bosworth. 2004. Intellectual property abuses: How should multinationals respond? *Long Range Plan.* **37**(5): 459–475.

Zaheer, A., N. Venkatraman. 1995. Relational governance as an interorganizational strategy: An empirical test of the role of trust in economic exchange. *Strateg. Manag. J.* **16**(5): 373–392.

Zhao, M. 2006. Conducting R&D in countries with weak intellectual property rights protection. *Management Sci.* **52**(8): 1185–1199.

Zhou, H., W. C. Benton. 2007. Supply chain practice and information sharing. *J. Oper. Manag.* **25**(6): 1348–1365.

Zhou, K. Z., D. Xu. 2012. How foreign firms curtail local supplier opportunism in China: Detailed contracts, centralized control, and relational governance. *J. Int. Bus. Stud.* **43**(7): 677–692.